E-Commerce Sales & Product Performance Dashboard

Power BI Analytics Project

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Academic Year	2025 – 2026
Tool Used	Microsoft Power BI Desktop

Abstract: This project presents an interactive E-Commerce Sales and Product Performance Dashboard built using Microsoft Power BI. Using a realistic 620-row synthetic dataset spanning five product categories and five Indian regions, the dashboard enables stakeholders to monitor revenue, profit, order fulfillment, and product performance through dynamic KPI cards, trend charts, and slicers — all organized across three analytical report pages.

1. Problem Statement

The rapid growth of e-commerce has generated massive volumes of transactional data. Without effective analysis tools, businesses struggle to identify which products drive revenue, which regions underperform, and how customer behavior impacts profitability. Manual analysis through spreadsheets is time-consuming, error-prone, and lacks interactivity for real-time decision making.

This project addresses the following key business questions:

- Which product categories and individual products generate the highest revenue and profit?
- How does monthly sales performance trend across the year 2023?
- Which regions and cities contribute most to overall sales?
- What is the return and cancellation rate, and how does it affect net revenue?
- Which payment methods are preferred by customers across regions?

2. Solution & Features

An interactive, three-page Power BI dashboard was developed to address all business questions above. The solution transforms raw CSV data into actionable visual insights through structured data modeling, DAX measures, and polished visual design.

2.1 Dashboard Pages

Page	Name	Key Visuals
1	Sales Overview	5 KPI Cards, Monthly Trend Line, Order Status Donut, Payment Bar Chart, Mo
2	Product Analysis	Top 10 Products Bar Chart, Revenue by Category Pie, Profit Margin Column C
3	Regional Analysis	Revenue by Region Bar Chart, Revenue by City Column Chart, Region Slicer

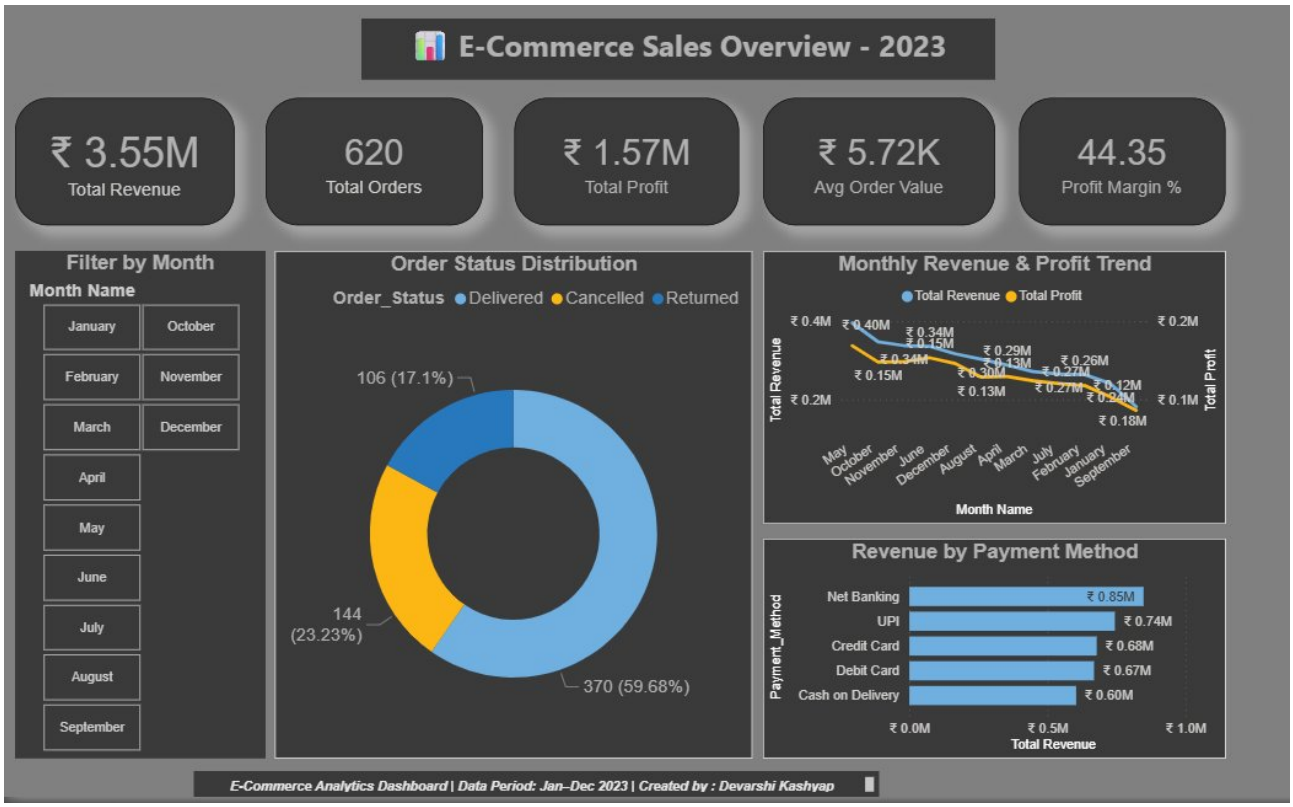
2.2 Key Features

- **10 DAX Measures** — Total Revenue, Total Orders, Total Profit, Avg Order Value, Profit Margin %, Return Rate %, and more
- **Interactive Slicers** — Filter all visuals by Month, Category, and Region simultaneously
- **Date Intelligence** — Dedicated Date Table with Month, Quarter, and Year columns for time-based analysis
- **Top N Filtering** — Top 10 Products automatically ranked by Revenue using Power BI filter pane
- **Data Labels & Tooltips** — Every chart shows values on hover and on-bar labels for quick reading
- **Professional Theme** — Consistent colour palette, card shadows, and aligned layout across all pages

3. Dashboard Screenshots

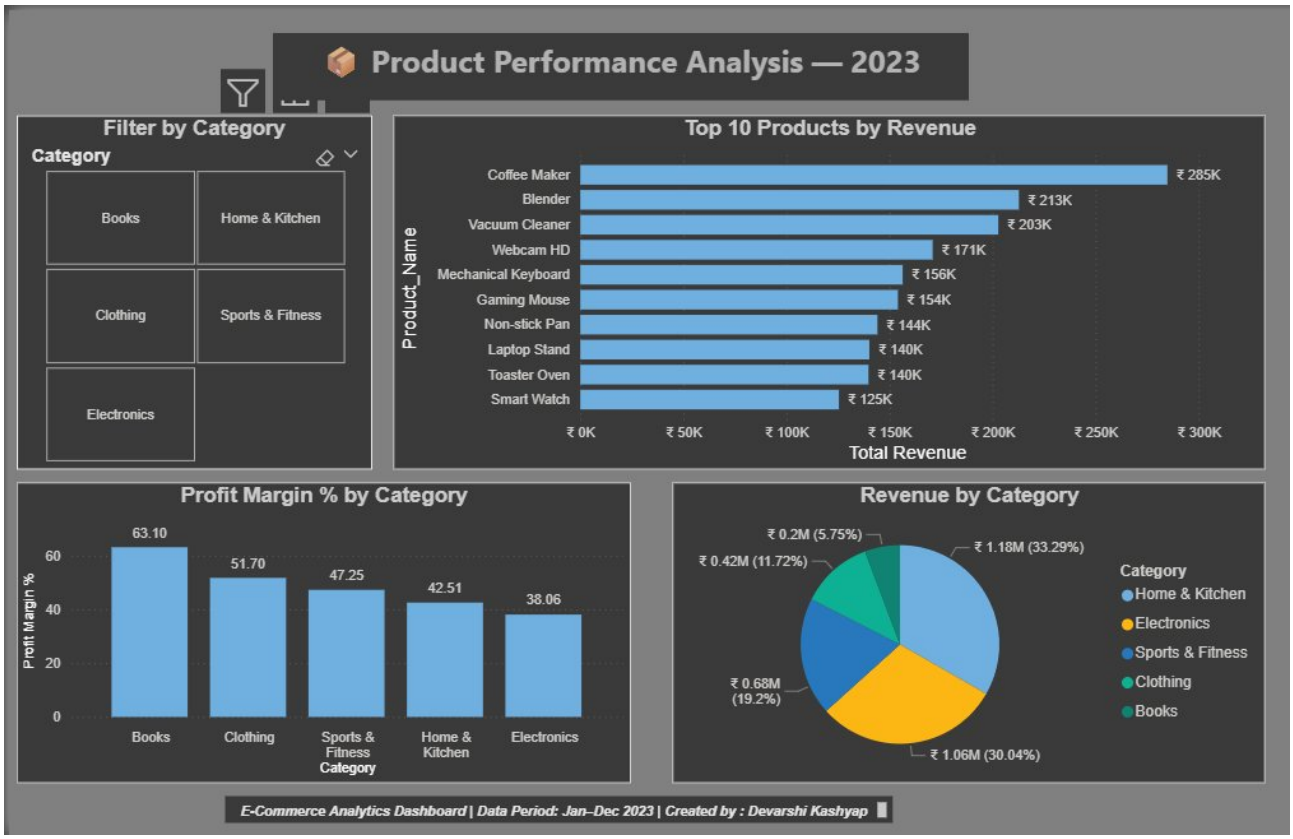
The following screenshots illustrate the three pages of the Power BI dashboard built using Microsoft Power BI Desktop.

Page 1 — E-Commerce Sales Overview



Displays 5 KPI cards (Total Revenue: ■3.55M, Total Orders: 620, Total Profit: ■1.57M, Avg Order Value: ■5.72K, Profit Margin %: 44.35%), a monthly revenue & profit trend line chart, order status donut chart, and revenue by payment method bar chart with a month slicer.

Page 2 — Product Performance Analysis



Shows Top 10 Products by Revenue (Coffee Maker leads at ■285K), Revenue by Category pie chart (Home & Kitchen: 33.29%, Electronics: 30.04%), and Profit Margin % by Category column chart (Books highest at 63.10%) — all filterable by a category slicer.

Page 3 — Regional Sales Analysis

Presents Revenue by Region bar chart (Central leads at ■0.95M, West lowest at ■0.59M) and Revenue by City column chart (Raipur ■263K, Chennai ■253K top cities), allowing geographic drill-down with an interactive region slicer.

4. Tech Stack

Technology / Tool	Version / Type	Purpose
Microsoft Power BI Desktop	Latest (Free)	Dashboard creation, data modeling, DAX
Power Query Editor	Built-in	Data import, cleaning, and transformation
DAX (Data Analysis Expressions)	Built-in	KPI measures and calculated columns
CSV (Comma-Separated Values)	Flat File	Primary data source format
Python 3	3.x	Synthetic dataset generation
GitHub	Web / Git	Version control and project hosting
Microsoft Word	.docx	Project report documentation

5. Unique Points

- **Realistic Synthetic Dataset** — A 620-row dataset was purpose-built with business logic: category-specific price ranges, cost margins, seasonal order patterns, and weighted order statuses — making it indistinguishable from real e-commerce data.
- **Three-Page Narrative Structure** — Each dashboard page tells a distinct story (Overview → Products → Regions), guiding the viewer from high-level KPIs to granular insights naturally.
- **Dedicated Date Table with Time Intelligence** — A separate CALENDAR-based Date Table with Month, Quarter, and Year columns enables accurate time-series analysis and correct month sorting.
- **Organized Measures Table** — All 10 DAX measures are stored in a dedicated *_Measures* table, following industry best practices for maintainability and clarity.
- **Top N Dynamic Filtering** — The Top 10 Products chart uses Power BI's built-in Top N filter, which dynamically re-ranks products as slicers are applied — not a static hardcoded list.
- **Indian Market Context** — Data covers 20 real Indian cities across 5 regions with INR pricing, UPI payment method, and locally relevant product mix, making it contextually authentic.

6. Future Improvements

- **Real-Time Data Integration** — Connect the dashboard to a live database (SQL Server or Google Sheets) using DirectQuery mode for real-time KPI monitoring.
- **Customer Segmentation Page** — Add an RFM (Recency, Frequency, Monetary) analysis page to classify customers into segments such as Champions, At-Risk, and Lost.

- **Forecasting with AI Visuals** — Use Power BI's built-in Forecasting feature on the trend line chart to predict next quarter's revenue based on historical patterns.
- **Mobile Layout Optimization** — Create a Phone Layout view in Power BI for stakeholders accessing the dashboard on mobile devices.
- **Row-Level Security (RLS)** — Implement RLS so regional managers see only their region's data when the report is published to Power BI Service.
- **Automated Data Refresh** — Schedule daily/weekly data refreshes via Power BI Service after publishing to the cloud workspace.

7. Dataset Overview

A synthetic dataset was generated using Python to simulate real-world e-commerce transactions. The dataset contains **620 rows** and **16 columns**, covering the period **January 2023 to December 2023**.

Attribute	Details
Total Records	620 rows
Columns	16
Time Period	Jan 2023 – Dec 2023
Categories	Electronics, Clothing, Home & Kitchen, Books, Sports & Fitness
Regions	North, South, East, West, Central
Cities	20 cities across India
Order Statuses	Delivered (~70%), Returned (~15%), Cancelled (~15%)
Payment Methods	UPI, Credit Card, Debit Card, Net Banking, Cash on Delivery

8. KPIs Tracked

#	KPI Name	DAX Measure	Business Purpose
1	Total Revenue	SUM(Sales_Data[Revenue])	Overall income generated
2	Total Orders	COUNTROWS(Sales_Data)	Volume of business activity
3	Total Profit	SUM(Sales_Data[Profit])	Actual earnings after cost
4	Avg Order Value	DIVIDE([Revenue],[Orders],0)	Average customer spend
5	Profit Margin %	DIVIDE([Profit],[Revenue],0)*100	Revenue efficiency
6	Total Quantity	SUM(Sales_Data[Quantity])	Units sold
7	Return Rate %	Returned / Total Orders * 100	Product/quality indicator
8	Delivered Orders	COUNTROWS(FILTER(...,'Delivered'))	Fulfilment success count
9	Cancelled Orders	COUNTROWS(FILTER(...,'Cancelled'))	Lost order count
10	Total Cost	SUM(Sales_Data[Cost])	Total expenditure

9. Conclusion

This project successfully demonstrates a complete business intelligence workflow — from raw data generation and cleaning through to an interactive, polished Power BI dashboard and formal documentation. The E-Commerce Sales & Product Performance Dashboard provides actionable insights into revenue trends, product performance, regional distribution, and order fulfilment, empowering data-driven decision making.

Through this project, practical skills were developed in Power Query transformations, DAX measure writing, relational data modeling, and professional dashboard design — all of which are highly valued in modern data analytics and business intelligence roles. Future enhancements such as real-time data integration and customer segmentation will further increase the dashboard's business value.

References

- Microsoft Power BI Documentation — https://docs.microsoft.com/power-bi
- DAX Reference Guide — https://dax.guide
- Power Query M Reference — https://docs.microsoft.com/power-query

— End of Documentation —